Difference lifts give unbounded-quotient NSS Jamison sequences

Candidate partial result for arXiv:1905.09592

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Status. Likely valid partial result. The argument produces Jamison sequences with unbounded consecutive quotients that are NSS for both group classes in source Problems 2.6 and 2.8. It does not settle either problem for every Jamison sequence.

1 Source problems and metadata

The source is C. Badea, V. Devinck, and S. Grivaux, *Escaping a neighborhood along a prescribed sequence in Lie groups and Banach algebras*, Canadian Math. Bull. **63**(3) (2020), 484–505, arXiv:1905.09592, DOI 10.4153/S0008439519000560 [1].

On PDF page 8, Problem 2.6 asks whether every Jamison sequence is NSS for the class of Banach–Lie groups.

> We do not know whether Corollary 2.5 can be extended to the class of Banach-Lie groups. See Remark 4.1 after the proof of Theorem 2.2 for a discussion of the difficulties that arise when considering Banach-Lie groups instead of (finite dimensional) Lie groups.
>
> **Problem 2.6.** Let $(n_k)_{k\geq 0}$ be a Jamison sequence with $n_0 = 1$. Is $(n_k)_{k\geq 0}$ NSS for the class of Banach-Lie groups?

Figure 1: Problem 2.6 in the source PDF.

Problem 2.8 asks the analogous question for groups with a minimal metric. The same page records the source's partial answer, Theorem 2.9, for sequences with bounded consecutive quotients.

2 Definitions

Let $Q = (q_k)_{k\geq 0}$ be a strictly increasing sequence of positive integers with $q_0 = 1$. For a topological group G with identity e, the pair (G, Q) is *NSS* if there is an identity neighborhood U such that

$$x^{q_k} \in U \text{ for all } k \geq 0 \quad \Longrightarrow \quad x = e.$$

A sequence is NSS for a class of groups if this holds for every group in that class.

For $\varepsilon > 0$, (Q, ε) is a *Jamison pair* if

$$\sup_{q\in Q} |\lambda^q - 1| \geq \varepsilon \qquad (\lambda \in \mathbb{T} \setminus \{1\}).$$

The supremum of the admissible ε is the Jamison constant.

Problem 2.8. Let $(n_k)_{k\geq 0}$ be a Jamison sequence with $n_0 = 1$. Is $(n_k)_{k\geq 0}$ NSS for the class of groups with a minimal metric?

The following result provides a partial answer to Problems 2.6 and 2.8 for sequences with bounded quotients.

Theorem 2.9 (Sequences with bounded quotients as NSS sequences)**.** *Let $(n_k)_{k\geq 0}$ be a sequence of integers with $n_0 = 1$ and*

$$\sup_{k\geq 0} \frac{n_{k+1}}{n_k} \leq c < +\infty.$$

Then $(n_k)_{k\geq 0}$ is NSS for the class of Banach-Lie groups and for the class of topological groups possessing a minimal metric.

Figure 2: Problem 2.8 and source Theorem 2.9.

3 Proof intuition

The source controls sequences whose own neighboring terms grow by a bounded factor. The key observation is that a selected exponent sequence can inherit NSS from a much better behaved sequence hidden among its differences. If $m = a - b$ and both x^a and x^b are close to the identity, then $x^m = (x^b)^{-1}x^a$ is also close. Thus an NSS witness for all the differences m_j becomes an NSS witness for the original selected exponents.

This permits large gaps: place two exponents A_j and $A_j + 2^j$ near each other, but place successive pairs extremely far apart. The selected sequence has unbounded quotients, while its differences contain the bounded-quotient sequence (2^j) covered by source Theorem 2.9. The original universal problems remain open because an arbitrary Jamison sequence need not visibly contain a known NSS sequence in its difference set.

4 The transfer theorem

Theorem 1 (difference-lift transfer)**.** *Let $Q \subset \mathbb{N}$ contain 1, and let $M = (m_j)_{j\geq 0}$ be a sequence of positive integers with $m_0 = 1$. Suppose that for every j there are $a_j, b_j \in Q$ such that*

$$a_j > b_j, \qquad m_j = a_j - b_j.$$

Then the following hold.

1. *For every class $\mathcal{C}$ of topological groups, if M is NSS for $\mathcal{C}$, then Q is NSS for $\mathcal{C}$.*

2. *If (M, ε) is a Jamison pair, then $(Q, \varepsilon/2)$ is a Jamison pair.*

Proof. Fix $G \in \mathcal{C}$ and let V be an identity neighborhood witnessing that M is NSS for G. Continuity of $(u, v) \mapsto u^{-1}v$ at (e, e) gives an identity neighborhood U such that

$$U^{-1}U \subseteq V.$$

If $x^q \in U$ for every $q \in Q$, then for every j,

$$x^{m_j} = x^{a_j - b_j} = (x^{b_j})^{-1}x^{a_j} \in U^{-1}U \subseteq V.$$

The NSS property of M gives $x = e$, proving the first assertion. Notice that no commutativity of G is needed: all factors are powers of the same element.

For the second assertion, let $\lambda \in \mathbb{T} \setminus \{1\}$ and fix $0 < \eta < \varepsilon$. Choose j with $|\lambda^{m_j} - 1| > \eta$. Since $|\lambda| = 1$,

$$\begin{aligned} \eta &< |\lambda^{a_j - b_j} - 1| = |\lambda^{a_j} - \lambda^{b_j}| \\ &\leq |\lambda^{a_j} - 1| + |\lambda^{b_j} - 1|. \end{aligned}$$

At least one of the two terms on the last line is greater than $\eta/2$, and both exponents belong to Q. Letting $\eta \uparrow \varepsilon$ gives the claimed lower bound $\varepsilon/2$. $\square$

Corollary 2. *If $Q - Q$ contains a positive sequence with bounded consecutive quotients and initial term 1, then Q is NSS for every Banach–Lie group and every group with a minimal metric.*

Proof. Source Theorem 2.9 [1] says that the difference sequence is NSS for both classes. Apply the transfer theorem. $\square$

5 An explicit unbounded-quotient sequence

Theorem 3. *Put*

$$A_j = 2^{2^{j+1}} \quad (j \geq 0), \qquad Q = \{1\} \cup \{A_j, A_j + 2^j : j \geq 0\},$$

and enumerate Q increasingly. Then:

1. *Q is a Jamison sequence with Jamison constant at least $\sqrt{3}/2$;*
2. *the consecutive quotients of Q are unbounded;*
3. *Q is NSS for every Banach–Lie group and every group with a minimal metric.*

Proof. The sequence starts

$$1, 4, 5, 16, 18, 256, 260, \ldots.$$

For every $j \geq 0$, the two members of the jth pair have difference 2^j. The sequence $(2^j)_{j\geq 0}$ has Jamison constant $\sqrt{3}$ [1]. Hence, for every $\lambda \neq 1$ on the unit circle and every $0 < \eta < \sqrt{3}$, some j satisfies

$$\eta < |\lambda^{2^j} - 1| = |\lambda^{A_j + 2^j} - \lambda^{A_j}| \leq |\lambda^{A_j + 2^j} - 1| + |\lambda^{A_j} - 1|.$$

One selected power is more than $\eta/2$ away from 1. Letting $\eta \uparrow \sqrt{3}$ proves (1).

Next, $A_{j+1} = A_j^2$ and $2^j \leq A_j$, so the quotient across successive pairs obeys

$$\frac{A_{j+1}}{A_j + 2^j} \geq \frac{A_j^2}{2A_j} = \frac{A_j}{2} \longrightarrow \infty.$$

The pair ordering is strict because $A_j + 2^j < A_{j+1}$, proving (2).

Finally, (2^j) has consecutive quotient two, so source Theorem 2.9 makes it NSS for both group classes. Since $(2^j) \subset Q - Q$, the transfer theorem proves (3). $\square$

6 Verification, limitations, and novelty

The proof uses only the identity-neighborhood inclusion $U^{-1}U \subset V$, the scalar triangle inequality, and source Theorem 2.9. No computation is used. The detailed adversarial audit is in `verification_report.md`.

This result does not answer Problems 2.6 or 2.8 for every Jamison sequence. It strictly enlarges the source's sufficient family by exhibiting a Jamison sequence with unbounded consecutive quotients that works for both classes.

Cheap run indexes and bounded searches on 19 July 2026 found no exact difference-transfer statement or later resolution. Searches used the exact problem wording together with "Jamison sequence", "NSS", "difference set", "unbounded quotients", "Banach–Lie", and "minimal metric". The result is therefore labeled *likely valid partial result*, not literature-certified novel.

Human-review recommendation. Promote after independently checking the neighborhood orientation and the explicit indexing. These are the only non-citation-dependent steps.

References

[1] C. Badea, V. Devinck, and S. Grivaux, *Escaping a neighborhood along a prescribed sequence in Lie groups and Banach algebras*, Canadian Math. Bull. **63**(3) (2020), 484–505; arXiv:1905.09592; DOI: 10.4153/S0008439519000560.